WEB DEVELOPMENT

YOUR FIRST STEP



From **zero** to your **first web page** — in one session.

Built for students who have never written a line of code

TODAY'S ROADMAP

1 · HTML

Web pages are made of HTML — and HTML is all about **structure**.

2 · TAGS

Meet the building blocks: `h1` `p` `img` `ul` `form` …

3 · FIRST PAGE

Create your very **first** HTML document from scratch.

4 · AI

It's 2026 — AI codes with us. How to use it **well**.

5 · EXAMPLE

A full page, explained line by line.

6 · QUIZ

Show what you know. No pressure!

7 · YOU CODE

Build your own resume in HTML.

8 · LET'S GO

Thank you — and time to code.

SECTION 01

WEB PAGES ARE HTML

Every page you've ever visited — Instagram, YouTube, your school portal — is a text document written in one language.



WHAT IS A WEB PAGE, REALLY?

A web page is a **plain text file** that a browser knows how to read.

That file is written in **HTML** — the language of the web.

When you open it, the browser **reads the HTML** and **draws** (renders) the page you see.

Type it once...

`HTML` → browser → page



*HTML is not a programming language — it has no logic, no math. It simply **describes** what is on the page.*

HTML = THE STRUCTURE

Think of a page like a **house**:



HTML

The **walls and rooms**. It decides what exists and where.

TODAY

CSS

The **paint and furniture**. Colors, fonts, layout.

LATER

JAVASCRIPT

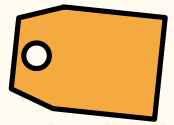
The **electricity**. Makes things react and move.

LATER

First we build a solid structure. Decoration comes later.

...AND HTML IS BASED ON TAGS

A **tag** is a keyword wrapped in angle brackets. Tags usually come in **pairs**:



```
<p>Hello!</p>
```

opening tag — *starts here*

content — *what is shown*

closing tag — *ends it (note the /)*

Some tags carry **attributes** — extra information:

```
<a href="https://github.com">my profile</a>
```

tag name — *what it is*

attribute — *a setting*

value — *in quotes*

content — *what's clicked*

TAGS THAT STAND ALONE

Some tags have **no content**, so they **close themselves**. We call them **void elements**.



```
<br>

<img src="cat.png" alt="A cat">
```

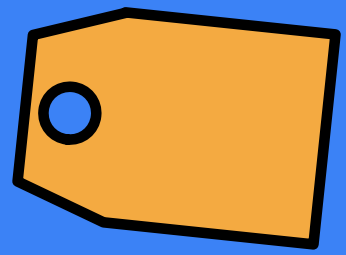
- `br` → a line break (like pressing Enter)
- `img` → an image, loaded from a file
- No closing `</br>` or `</img>` — there is nothing inside them

*Rule of thumb: if a tag **contains** something, it needs a closing tag. If it **is** something, it doesn't.*

SECTION 02

MEET THE TAGS

A handful of tags is enough to build a real page.
Here are the ones you'll use every single day.



Headings — `h1` to `h6`

```
<h1>My Web Page</h1>
<h2>About me</h2>
<h3>My studies</h3>
<h4>First year</h4>
<h5>Notes</h5>
<h6>Tiny detail</h6>
```



Use **one** `h1` per page. Then `h2` for main sections, `h3` … for subsections. This is the page outline.

*Headings are not for making text **big** — they give the page meaning. Search engines and screen readers rely on them.*

PARAGRAPHS & LINE BREAKS – `p`, `br`

```
<p>I am a student.</p>

<p>I study computer science.<br>
This is on a new line, same paragraph.</p>
```

I am a student.

I study computer science.
This is on a new line, same paragraph.

- **`p`** → a paragraph. The browser adds space before and after.
- **`br`** → a line break *inside* a paragraph.

Don't use lots of `<br>` to fake spacing — use real tags for real structure.

IMAGES – `img`

```
<img src="photo.jpg" alt="My portrait">
```

- **`src`** → the path to the image file
- **`alt`** → text shown if the image fails, and read aloud by screen readers



`alt` is not optional — it is how blind users "see" your image.

A good `alt` describes the image: `alt="Cat sleeping on a laptop"`, not `alt="cat"`.

LISTS – `ul`, `ol`, `li`

Unordered (bullets) and **ordered** (numbers):

```
<ul>
  <li>HTML</li>
  <li>CSS</li>
</ul>

<ol>
  <li>Open the editor</li>
  <li>Write the code</li>
</ol>
```

- HTML
- CSS

1. Open the editor
2. Write the code

Every item lives in its own `<li>`. Lists can even nest inside each other.

FORMS — COLLECTING INPUT

```
<form>
  <label for="name">Your name:</label>
  <input type="text" id="name">

  <label for="msg">Message:</label>
  <textarea id="msg"></textarea>

  <button type="submit">Send</button>
</form>
```

`form`
Wraps the whole input area.

`label`
Describes a field. Links to it with `for`.

`input`
The field itself. `type` = text, email, checkbox…

`button`
Sends the form.

YOUR CHEAT SHEET (1/2)

TAG	WHAT IT DOES	EXAMPLE
`h1` … `h6`	Headings (1 = biggest)	`<h1>Hi</h1>`
`p`	Paragraph of text	`<p>Hello</p>`
`br`	Line break (self-closing)	`line one line two`
`img`	Image (self-closing)	`<img src="a.png" alt="a">`
`ul` / `ol`	Unordered / ordered list	`<ul><li>x</li></ul>`
`li`	A list item	`<li>Item</li>`

Spotted the pattern? Some tags **wrap** content, some **stand alone.**

YOUR CHEAT SHEET (2/2)

TAG	WHAT IT DOES	EXAMPLE
`form`	Input area	`<form>…</form>`
`input`	A form field	`<input type="text">`
`label`	Describes a field	`<label for="x">X</label>`
`button`	Clickable button	`<button>Send</button>`
`a`	A link	`<a href="…">go</a>`

You now know more than enough HTML to build a real page!

SECTION 03

YOUR FIRST HTML DOCUMENT

Every page on the web starts with the same skeleton. Learn it once, use it forever.



THE SKELETON OF EVERY PAGE

```html
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="utf-8">
    <title>My first page</title>
  </head>
  <body>
    <h1>Hello, world!</h1>
    <p>This is my very first page.</p>
  </body>
</html>
```

- `<!DOCTYPE html>` → "this is a modern HTML page"
- `<html>` → the whole document
- `<head>` → invisible info (title, encoding, links)
- `<title>` → text in the browser tab
- `<body>` → everything you actually see

The `head` is the page's brain (info). The `body` is its face (visible content).

CREATE IT IN 4 STEPS

1. **Make a folder** for your project, e.g. `my-site`.
2. **Create a file** called `index.html`. The `.html` ending tells the computer it's a web page.
3. **Write** the skeleton, then your content, and **save**.
4. **Open it** — double-click the file, or drag it into a browser. Refresh (`F5`) to see your changes.

EDITOR

VS Code, or any text editor.
Not Word!

PREVIEW

Your browser **is** the preview.
No install needed.

INDEX.HTML

The name browsers use for the home page.

See your mistakes — the browser developer tools

- Press **F12** (or right-click → *Inspect*) to open **DevTools**.
- The **Elements** tab shows the HTML as the browser understood it.
- Red errors in the **Console** help you find problems.



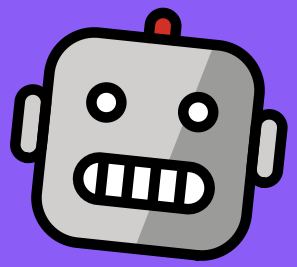
This is your **best friend** when something looks wrong.

> *Professional developers keep DevTools open **all day long**. Start the habit now.*

SECTION 04

CODING WITH AI IN 2026

You will never code alone again. AI is a co-pilot — but you are the pilot.



FREE AI CODING TOOLS YOU CAN USE TODAY

OPENCODE

An open-source AI coding agent in your terminal. Great for learning by asking questions.

CLAUDE

Excellent at explaining code and fixing mistakes you paste in.

GEMINI

Google's assistant — handy for quick snippets and debugging.

CODEX

OpenAI's coding assistant, built into many editors.

GITHUB COPILOT

Free for students — autocompletes as you type.

CURSOR

An editor with AI built right in.

Free tiers change fast — check what your school provides, and always use your own account.

BE THE PILOT, NOT THE PASSENGER

DO 🎯

- Ask AI to **explain** code line by line
- Ask **"why"**, not just "give me the answer"
- Run everything in the browser and test
- Rewrite the code in your own words

DON'T ✖

- Paste code you don't understand
- Trust it blindly — it **will** be wrong sometimes
- Give away passwords, keys or private data
- Let it do your homework (you're learning, not it)

Good prompt: "Show me the HTML for a page with my name as a title and a list of 3 skills, and explain every tag."

SECTION 05

A FULL EXAMPLE

One small page that uses everything we've seen — then we read it together.



THE COMPLETE DOCUMENT (1/2)

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="utf-8">
    <title>Sara's page</title>
  </head>
  <body>
    <!-- notes for humans; the browser ignores this -->

    <h1>Sara Benali</h1>
    <img src="sara.jpg" alt="Portrait of Sara">
```

...AND THE BODY CONTINUES (2/2)

```
    <h2>About me</h2>
    <p>First-year student.<br>I love maths and music.</p>

    <h2>Skills</h2>
    <ul>
      <li>HTML</li>
      <li>Problem solving</li>
    </ul>
    <h2>Contact</h2>
    <form>
      <label for="email">Email:</label>
      <input type="email" id="email">
      <button type="submit">Send</button>
    </form>
  </body>
</html>
```

READING IT LINE BY LINE

LINES	WHAT IT IS	WHY IT MATTERS
`<!DOCTYPE html>`	The declaration	Tells the browser to use modern standards
`<html lang="en">`	The root element	`lang` helps screen readers & translators
`<head> … </head>`	Invisible info	Not shown, but essential
`<meta charset="utf-8">`	Text encoding	So accents like `é` display correctly
`<title>`	Tab title	Shows in the browser tab & search results
`<body> … </body>`	Visible content	Everything the visitor sees
`<!-- … -->`	A comment	Notes for you, hidden from visitors

READING IT LINE BY LINE (PART 2)

TAG USED	ROLE IN THE PAGE
`<h1>Sara Benali</h1>`	The main title — the person's name
`<img src="sara.jpg" alt="Portrait of Sara">`	Adds a picture, with a description
`<h2>About me</h2>`	Starts a new section
`<p> …   … </p>`	A paragraph with a line break inside
`<ul><li>…</li></ul>`	A bullet list of skills
`<form>` + `<input>` + `<button>`	A way for visitors to contact Sara

STRUCTURE

Notice how headings create an outline: `h1` then several `h2`.

NO CSS, NO JS

It's ugly — and that's fine. The **meaning** is already all there.

SECTION 06

QUIZ TIME!

Five quick questions. Shout out your answer — or write it down. No grades, I promise.



QUESTION 1

1 What does HTML stand for?

A. Hyperlinks and Text Markup Language

B. HyperText Markup Language

C. Home Tool Markup Language

2 Which tag makes the biggest heading? **A.** `<h6>` **B.** `<head>` **C.** `<h1>`

3 Which tag is **self-closing** (void)? **A.** `<br>` **B.** `<p>` **C.** `<li>`

QUESTION 2

4 Where do you put the content that visitors actually **see**? **A.** inside `<head>` **B.** inside `<body>` **C.** inside `<title>`

5 What is the `alt` attribute for on an image? **A.** To set the image size **B.** To describe the image **C.** To change its colour

6 Should you paste AI code you don't understand into your project? **A.** Yes, always **B.** Only if it runs **C.** No — understand it first

ANSWERS

1. **B — HyperText Markup Language.** "Markup" means we *mark up* content with tags.

2. **C —** `<h1>`. Headings go from `h1` (biggest) to `h6` (smallest).

3. **A —** `<br>`. It has no content, so it doesn't need a closing tag.

4. **B — inside** `<body>`. The `head` holds invisible info.

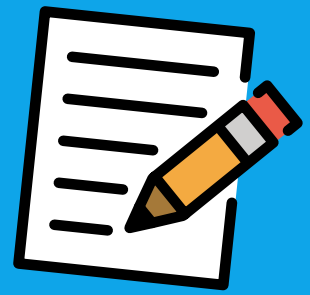
5. **B — to describe the image.** It also helps screen readers and shows when an image fails to load.

6. **C — no, understand it first.** AI is your co-pilot, not your replacement.

SECTION 07

YOUR TURN: BUILD YOUR RESUME

One HTML file. No CSS. No JavaScript. Just you, your structure, and your story.



YOUR MISSION

Create a file called `resume.html` that presents **your resume** in pure HTML.

Rules:

- ☑ Only **HTML** — no CSS, no JavaScript, no frameworks
- ☑ One single file
- ☑ It must open and look correct in a browser
- ☑ Use your own information (or invent a character!)
- ☑ Test it before finishing

WHY NO CSS?

Because today is about **structure and meaning**. Ugly is fine. A solid skeleton is what matters — the pretty version comes next lesson.

TIP

Copy the skeleton from section 3 and grow it. Add sections one at a time and refresh often.

YOUR PAGE MUST CONTAIN...

STRUCTURE

`<!DOCTYPE>`, `html`, `head`, `body`, and a `title`.

YOUR NAME

As an `h1`, plus `h2` headings for each section.

A PHOTO

An `img` with a real `alt` description (a placeholder is OK).

A SUMMARY

A `p` paragraph introducing yourself.

TWO LISTS

A `ul` for skills, an `ol` for education or experience.

A CONTACT FORM

`form` + `label` + `input` + `button`.

A LINE BREAK

Somewhere sensible, using `br`.

COMMENTS

Separate your sections with `<!-- ... -->` notes.

STARTER SKELETON (1/2)

```html
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="utf-8">
    <title>My Resume</title>
  </head>
  <body>

    <!-- Header -->
    <h1>Your Name</h1>
    <img src="me.jpg" alt="Photo of Your Name">
```

STARTER SKELETON (2/2)

```
    <!-- About -->
    <h2>About me</h2>
    <p>Write one or two sentences about yourself here.</p>

    <!-- Skills -->
    <h2>Skills</h2>
    <ul>
      <li>Your first skill</li>
      <li>Your second skill</li>
    </ul>

    <!-- Next: Education, Experience, Contact... -->

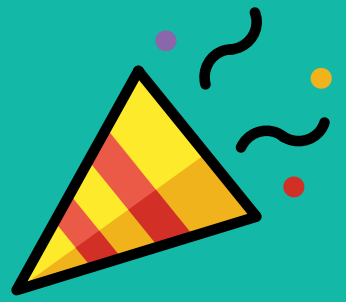
  </body>
</html>
```

Now add Education, Experience, Languages, and a Contact form. Make it yours!

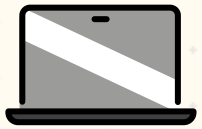
SECTION 08

THANK YOU!

You just learned the language of the entire web.
That's a big deal.



NOW LET'S CODE


Open your editor. Create `resume.html`.

Ask AI when you're stuck — then **make it yours**.

Questions? Hands up. Let's build.